

The empirical recurrence for OEIS A250514: recovering a conjecture that is not written down

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Abstract

OEIS A250514 records “Empirical recurrence of order 50”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 52$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 50, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 50 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A250514 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with nondecreasing $\min(x(i,j), x(i,j-1))$ in the i direction and nondecreasing $\min(x(i,j), x(i-1,j))$ in the j direction.”. It has offset 1 and begins

5996, 205022, 5941982, 166583429, 4464289109, 117638074883, 3049397341176, 78567969110807, ...

The entry states:

Empirical recurrence of order 50 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:29:12, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 52$ of the 256, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 52$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 104$ exact terms of the model returns order 50 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{50} c_i a(n-i),$$

c_1	34	c_{14}	−43422949768128	c_{27}	−42900747845081720064	c_{40}	−255739745717945892
c_2	261	c_{15}	−275515120052537	c_{28}	−4352980615800170624	c_{41}	2180464618996582318
c_3	−16431	c_{16}	667645665201026	c_{29}	121408208416569476352	c_{42}	1434771469700982374
c_4	28001	c_{17}	4098596546440968	c_{30}	31504815035795370496	c_{43}	−821698534866419712
c_5	2740249	c_{18}	−7178844486178680	c_{31}	−264430483901619320320	c_{44}	−560836751559512555
c_6	−10565886	c_{19}	−45605243448275052	c_{32}	−100328174888589462528	c_{45}	2038860441280158103
c_7	−241401307	c_{20}	54060070980323448	c_{33}	443126456668247224320	c_{46}	1438077841299210240
c_8	1071602699	c_{21}	381668926694121592	c_{34}	208305168489270945792	c_{47}	−29385505535353159
c_9	13345125148	c_{22}	−279276792495587856	c_{35}	−569277525167457398784	c_{48}	−21473536319579750
c_{10}	−57863788103	c_{23}	−2416454789280863680	c_{36}	−306464037826789122048	c_{49}	18103250350138982
c_{11}	−505706609529	c_{24}	913586694625426208	c_{37}	556103049695904006144	c_{50}	13866319417127731
c_{12}	1943503404355	c_{25}	11641743624327594848	c_{38}	327863698842886078464		
c_{13}	13766291794097	c_{26}	−1236120346220230592	c_{39}	−407101841515756781568		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 50, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $50 + 4$ terms removed returns order 50 as well, so $\deg q_{\min} = 50 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $50 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 50 that this sequence satisfies — and that family turns out to have exactly one member.

References

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